Team Responsibility Matrix

Project: Remote Patient Vital Signs Monitoring System / RPM IoMT Patient Monitoring Node

Repository: `RPMIOMTproject`, branch `main`, HEAD `f67c13a`

Purpose: divide substantial technical ownership for the HOD individual discussions, define interfaces and handoffs, and prevent "everyone owns everything" answers.

Authorship warning: this is a **proposed** responsibility matrix. Names are placeholders because no names were supplied. It does not prove that any placeholder author wrote a file. The team must confirm the real allocation and revise this matrix if needed. Until confirmed, use **"the team implemented"** for source-backed shared work.

1. Ownership rules

1. **Every member owns a technical domain, not a slide deck.** Each area includes files, behavior, failure modes, tests and evidence.
2. **Ownership means responsibility to explain and defend**, not exclusive authorship.
3. **Cross-module behavior is shared.** No member may "hand off" the temperature semantics, null policy or evidence classification.
4. **Names are placeholders.** `[MEMBER 1 NAME]` … `[MEMBER 5 NAME]` must be replaced by the team after the real allocation is agreed.
5. **Source does not establish personal authorship.** A commit history or a team statement is needed before making a personal authorship claim.
6. **One decision owner per cross-cutting rule.** The owner coordinates changes; implementation remains shared.
7. **No production source was changed by this documentation task.** The matrix describes the current working tree, including prior uncommitted repairs.

2. The five substantial ownership areas

Area	Member	Primary technical	Deep files / artefacts	Evidence the member	Main dependencie

		ownership		must be able to show	s
A. Firmware and system integration	[MEMBER 1 NAME]	Boot/loop scheduling; I2C integration; sensor state machine; JSON serialization; OLED; SOS/fall; hot-plug; build configuration	src/main.cpp; platformio.ini; boards/esp32-s3-devkitc-1-n16r8.json; src/tmp117_protocol.h (consumer)	Main and diagnostic build results; firmware walkthrough; status/null paths; SOS/fall logic; hardware limitations	Hardware truth from Area B; schema consumed by Areas C/D; safety rules from Area E
B. Hardware, sensors and measurement reliability	[MEMBER 2 NAME]	Wiring/bus topology; TMP117 protocol and raw diagnostic; MAX30102 quality guards; MPU scaling/calibration; physical validation method	src/tmp117_protocol.h; src/sensor_math.h; diagnostics/tmp117_raw/; wiring-guide.html; README.md hardware sections	TMP117 raw conversion/identity/config; diagnostic build; MAX/MPU explanation; prior-observation labels; reference-test procedure	Firmware integration from Area A; bridge/dashboard revalidation from C/D; policy from E
C. Communication and backend	[MEMBER 3 NAME]	Serial discovery/reader; framing; duplicate and duplicate-key handling; strict schema/status/range validation; freshness;	telemetry_bridge.py; tests/test_telemetry.py; tests/fixtures.py; README.md telemetry/API sections	Parser/validator walkthrough; rejection cases; stale/no-replay test; API map; storage-error behavior; 62-test	Firmware contract from A; raw rules from B/E; storage/dashboard expectations from D

Area	Member	Primary technical ownership	Deep files / artefacts	Evidence the member must be able to show	Main dependencies
		SSE/HTTP APIs; storage handoff; local security posture		evidence	
D. Dashboard and storage	[MEMBER 4 NAME]	Exact labels; live DOM/SSE behavior; null/gap-aware chart and CSV; SQLite schema/migration/retention; verified wrist storage; device isolation	index.html; assets/dashboard.js; assets/dashboard_core.js; assets/dashboard.css; telemetry_store.py; tests/dashboard_core.test.js; tests/test_history.py	Label/display walkthrough; 10 Node tests; history tests; gap/CSV behavior; schema/migration; browser/E2E limits	Backend API/SSE from C; safety/contact policy from E; producer fields from A
E. Analytics, safety and quality	[MEMBER 5 NAME]	Temperature/fever policy; baseline statistics; deterministic alert engine; audit/evidence classification; quality gates	temperature_safety.py; personal_baseline.py; ai_decision_engine.py; tests/test_temperature_safety.py; tests/test_baseline.py; tests/test_decision_engine.py; audit documents	Exact semantics; baseline formula/qualification; deterministic thresholds; C-01–C-14/O-01–O-07 status; evidence labels; test/build limits	Measurements from B/D; packet fields from A/C; storage/baseline API from D

2.1 Why these are substantial

- **Area A** owns the integration that makes the whole repository a system rather than isolated sensor sketches.
- **Area B** owns the raw measurement chain and the physical-verification gap; it is not "just wiring."
- **Area C** owns the trust boundary that rejects malformed or contradictory packets and prevents stale replay.
- **Area D** owns the clinical display/history semantics: exact labels, nulls, gaps, CSV and verified-contact-only storage.
- **Area E** owns the safety policy, statistical baseline, deterministic alerting and evidence discipline.

No area is merely "documentation" or "presentation." Each has code and testable behavior.

3. Domain charters

3.1 Area A — Firmware and system integration ([MEMBER 1 NAME])

Owns the explanation of:

- `setup()` and `loop()` execution order.
- Primary `Wire` (GPIO 8/9, 50 kHz) and dedicated `Wire1` (GPIO 17/18, 400 kHz).
- Sensor initialization and status propagation: `valid`, `warming_up`, `low_quality`, `stale`, `sensor_error`, `not_available`.
- `outputTelemetry()` schema-2 JSON, `[JSON]` prefix, `115200` baud, `rpm-2.1.0`.
- Actual TMP117 versus clinical wrist fields at the firmware boundary.
- SOS GPIO 5 active-low path, 300 ms debounce, 15 s auto-clear.
- Fall candidate: free fall `<0.6 g`, impact `>2.5 g`, 60–600 ms, 6 s active.
- Hot-plug probes and 3 s retry.
- OLED compact-text limitation.
- Build flags: `RPM_ENABLE_OPTIONAL_SHT40=0`, `RPM_ENABLE_OPTIONAL_MLX90614=0`.

Must not claim: physical sensor/OLED/SOS validation; TMP117 accuracy; wrist/core temperature.

Primary handoff: packet contract and statuses to Area C; actual/wrist semantics to Areas D/E; raw hardware questions to Area B.

3.2 Area B — Hardware, sensors and measurement reliability ([MEMBER 2 NAME])

Owns the explanation of:

- Installed inventory and explicit non-installed modules.
- Power/ground/address strapping, primary/IMU buses and pull-ups.
- TMP117 `WHO_AM_I=0x0117`, configuration `0x01A0`, `DATA_READY`, offset, signed conversion `0.0078125 °C/LSB`.
- Raw diagnostic: address scan `0x48–0x4B`, config readback, raw/failure/stuck counters.
- Rejection of `0x0000/0xFFFF`, `-55..150 °C` actual range, `>5 °C` confirmation.
- MAX30102 DC/AC/PI guards and HR/SpO2 limits.
- MPU ±8 g/±250 dps, `4096 LSB/g`, `131 LSB/dps`, single-orientation bias limits.
- Reference-measurement and contact-verification protocols.

Must not claim: accuracy without a reference; prior `0x49`/raw observations as reproduced; SHT40/MLX as installed; wrist/core temperature from TMP117.

Primary handoff: raw/electrical evidence and caveats to Areas A/E; hardware status expectations to Areas C/D.

3.3 Area C — Communication and backend ([MEMBER 3 NAME])

Owns the explanation of:

- Serial discovery (`find_serial_port()` unique USB VID), reader thread, 115200 baud, buffer framing.
- `DuplicateSuppressor` 1.5 s versus duplicate JSON-key rejection.

- parse_packet() and _validate_packet(); [JSON] producer convention.
- _validate_measurement(), _validate_temperature(), _validate_fever(), _validate_imu().
- Sensor/active/active_max consistency (active_max = 3).
- snapshot() freshness at 5 s and no stale replay.
- SSE /api/stream queue 8/drop-oldest.
- HTTP routes and status codes; 127.0.0.1 binding; POST baseline origin/size/device checks.
- Storage handoff and nonfatal storage_error.

Must not claim: real serial/load/security validation; clinical accuracy; full authentication/TLS.

Primary handoff: accepted packet shape/API to Area D; rejection reasons and schema limits to Areas A/D/E.

3.4 Area D — Dashboard and storage ([MEMBER 4 NAME])

Owns the explanation of:

- Four active metrics and exact labels **Wrist Skin Temperature (°C)** / **TMP117 Sensor Temperature (°C)**.
- DashboardCore.metricStatus(), metricValue(), temperatureAssessment(), gapAwarePath(), csv().
- Live SSE consumption, _received_at duplicate-render guard, history request token.
- Gap-aware chart; blank CSV cells; active columns only.
- SQLite schema/migration/retention; measurements() verified-contact-only wrist; history() buckets/gaps/device isolation.
- Core/fever display with exact message.
- Browser/accessibility/security evidence limits.

Must not claim: full browser E2E/WCAG/security audit; wrist currently populated; actual TMP117 in history/CSV.

Primary handoff: storage/API expectations to Area C; baseline/event display needs to Area E; label/evidence gaps to all.

3.5 Area E — Analytics, safety and quality ([MEMBER 5 NAME])

Owns the explanation of:

- `temperature_safety.py` actual/wrist/core/fever rules and exact message.
- Raw tolerance/stuck/range checks; no fixed offset.
- Baseline: 7 days/30 minutes/day, minute means, median/MAD, noise floors, freeze, partial/ready states, deviation rules.
- Deterministic engine: RED/YELLOW/GREEN/UNKNOWN, HR 50/120, SpO2 90, missing optical data.
- No trained model/LLM.
- Audit/cleanup/next-step evidence: C-01–C-14 closed in software; O-01–O-07 open.
- Test/build evidence labels and release gates.

Must not claim: clinical validation, core/fever capability, AI model, all audit items closed, hardware accuracy.

Primary handoff: policy and evidence rules to Areas A–D; baseline expectations to Area D; hardware/clinical open items to all.

4. Cross-cutting shared ownership

These topics are **shared by all five members** and must not be left to one person:

Shared topic	Why everyone needs it	Decision coordinator
Actual TMP117 ≠ clinical wrist ≠ core	Prevents the project's most serious semantic error	Area E
Null instead of zero/stale	Applies to firmware, API, store, dashboard, CSV	Area C
Exact clinical labels/message	Appears in JSON, dashboard, OLED shorthand, terminal	Area D/E
`active_max = 3`; SHT40/MLX absent	Inventory claim repeated across modules	Area A/B
Evidence classification	Prevents tests turning into hardware/clinical claims	Area E
No core/fever invention	Safety boundary for all outputs	Area E

Shared topic	Why everyone needs it	Decision coordinator
Schema-2 contract	Producer/consumer change control	Area C
Hardware validation gap	Must be stated by every member	Area B
No personal authorship without evidence	Protects team honesty	All

4.1 Decision rights

Decision	Coordinator	Required consultation
Firmware packet/field/status change	[MEMBER 1 NAME]	Areas C, D, E
Sensor range/raw/protocol/hardware claim	[MEMBER 2 NAME]	Areas A, E
Validation/API/SSE/security behavior	[MEMBER 3 NAME]	Areas A, D
Storage schema/label/history/CSV behavior	[MEMBER 4 NAME]	Areas C, E
Temperature/baseline/alert policy	[MEMBER 5 NAME]	Areas A–D (all implement)
Evidence classification in documents	[MEMBER 5 NAME]	All source owners
Final HOD claim wording	Team consensus	All five

A coordinator does not own every line; they own consistency of the decision and the evidence trail.

5. Interface handoffs

5.1 Handoff table

From → To	Artefact	Acceptance condition	Failure mode if skipped
A → B	Firmware TMP117 read/config/status	Raw issue can be reproduced without	Hardware debugging blames software or

From → To	Artefact	Acceptance condition	Failure mode if skipped
	behavior; raw diagnostic need	main firmware ambiguity	vice versa
B → A	Confirmed electrical/identity/range constraints	Address/WHO_AM_I/config/raw expectations align	Firmware accepts wrong device or rejects valid hardware
A → C	Schema-2 sample packet and status semantics	parse_packet() + _validate_packet() accept the intended null/valid pattern	Live dashboard empty or invalid packets
B → C	Raw/stuck/range/tolerance rules	Bridge enforces independent equivalent checks	A stuck raw value reaches UI/history
C → D	SSE/API payloads; history/baseline/events routes	Dashboard can render connected/disconnected/error states	UI shows stale/blank incorrectly
D → C	Required fields/labels/gap behavior	API returns data the dashboard can validate/display	Label/value mismatch
D → E	Stored minute means/events/baseline API	Baseline can compute qualified days and comparisons	Baseline uses raw/unverified data
E → A–D	Exact temperature/fever/baseline policy	Every layer has matching tests and labels	One layer invents core/wrist/fever
E → all	Evidence tags and open-item status	Claims match source/test/build/HW reality	Tests presented as hardware/clinical proof
All → E	Test/build/hardware evidence with dates	Audit document reflects current state	Stale claims in viva

5.2 Handoff quality rules

1. State the exact source path and function, not just "the backend."

2. State the evidence tag and date for every test/build/hardware claim.

3. **State the failure mode** the receiver must handle (null, stale, error, gap).

4. **Preserve the distinction** between actual sensor temperature and clinical wrist.

5. **Never hand off a fabricated "working" claim**; hand off the limitation and the next test.

6. Evidence ownership

Evidence	Owner of explanation	Who may be asked
Main firmware build success and sizes	[MEMBER 1 NAME]	All
TMP117 diagnostic build/raw procedure	[MEMBER 2 NAME]	All
src/tmp117_protocol.h vectors	[MEMBER 2 NAME]	1, 5
Bridge validation/tests	[MEMBER 3 NAME]	All
Dashboard Node tests	[MEMBER 4 NAME]	3, 5
History Python tests	[MEMBER 4 NAME]	3, 5
Temperature/fever tests	[MEMBER 5 NAME]	All
Baseline/engine tests	[MEMBER 5 NAME]	4
Audit register/cleanup/next steps	[MEMBER 5 NAME]	All
Prior hardware observations	[MEMBER 2 NAME]	1, 5
Hardware validation gap	[MEMBER 2 NAME]	All

6.1 Exact current evidence summary

- `python -m pytest -q` → **62 passed** (fresh 2026-10-29).
- `node --test tests/dashboard_core.test.js` → **10 passed** (fresh 2026-10-29).
- Ruff, Python compilation and JavaScript syntax checks → passed.
- `pio run -e esp32-s3-devkitc-1` → **SUCCESS**; RAM 21,424/327,680 bytes; flash 326,001/6,553,600 bytes.
- `pio run -d diagnostics/tmp117_raw` → **SUCCESS**; RAM 18,916/327,680 bytes; flash 282,377/6,553,600 bytes.
- No fresh hardware upload, serial capture, reference thermometer, contact verification or assembled end-to-end test.

Prior hardware observations (0x49, raw 0x0ED3 → 29.6484375 °C, raw 0xFFFF → -0.0078125 °C, diagnostic 0x0DA7 → 27.30469 °C, 60 reads/0 failures/0 stuck patterns) remain **RECORDED HW OBS — not reproduced; HARDWARE VERIFICATION REQUIRED.**

7. Substantial ownership coverage check

Use this table before the viva to ensure no member is presenting only superficial material.

Member	Code they can trace	Debug scenario they can lead	Test/build evidence they can explain	Cross-module question they can answer
1	src/main.cpp boot/loop/output/OLED	SOS/fall/stale/hot-plug	Main + diagnostic builds, Python/Node counts	Why actual TMP117 is not wrist; how bridge consumes packet
2	TMP117 header/diagnostic; MAX/MPU init	TMP raw anomaly/identity; MAX quality; MPU calibration	Vector/compile-time tests; diagnostic build	How bridge/dashboard recheck raw/range
3	Serial reader/validators/SSE/APIs	malformed packet; duplicate/stale; storage error; no port	62 Python suite; schema rejection tests	How dashboard gets fresh/history data; why unverified wrist rejected
4	Dashboard core/DOM; SQLite store	blank wrist; chart gap; CSV zero bug; migration	10 Node; history tests	How store/baseline handle unverified wrist; exact labels
5	safety/baseline/engine	conflation; baseline not ready; UNKNOWN; AI-name	62 Python; temperature/baseline/engine tests; C/O audit items	Where policy is enforced across layers; baseline/storage interaction

If a member cannot fill a row, the allocation is not yet substantial enough; reassign or deepen the preparation before the HOD meeting.

8. Conflict and error protocol

1. **Source wins over memory.** Check the file/function before repeating a claim.
2. **Tests win over intuition**, but only for the behavior they exercise.
3. **Hardware evidence wins over both**, once captured with a dated protocol.
4. **A safety concern stops the claim.** If actual sensor temperature is being called wrist/core, the answer is wrong regardless of UI convenience.
5. **Do not fix the demo by weakening validation.** Fix the producer or version the contract.
6. **Do not invent an owner.** If authorship is unknown, say the team implemented it and identify the source path.
7. **Do not use prior hardware observations as current validation.**
8. **Escalate unresolved semantic conflicts to Area E**, and unresolved packet/API conflicts to Area C, before the viva.

9. Red lines for every member

- Never present actual TMP117 temperature as clinical wrist or core temperature.
- Never claim core estimation or fever screening; core is null/not_available, fever message is exact.
- Never present SHT40/MLX90614 as installed/tested.
- Never claim sensor accuracy without a reference measurement.
- Never claim a trained AI model/LLM exists.
- Never claim the physical device/OLED/SOS/fall was validated in this pass.
- Never claim all audit items are closed.
- Never claim personal authorship without team confirmation.
- Never hide a gap as zero or a line through missing history.
- Never call this prototype a clinically validated diagnostic/emergency system.

10. Team sign-off checklist

- ☐ Real names replace all [MEMBER n NAME] placeholders.
- ☐ Each member confirms the files/behaviors they can defend.
- ☐ Each member can answer the two cross-module questions in their document.
- ☐ The team agrees on exact labels/message/range semantics.
- ☐ The team agrees on the evidence tags and current test/build counts.
- ☐ The team agrees that prior TMP117 observations are not reproduced.
- ☐ The team agrees that SHT40/MLX are not installed.
- ☐ The team agrees that the engine is deterministic, not AI/LLM.
- ☐ The team agrees on the hardware/clinical validation gaps.
- ☐ The team agrees no production source was changed by this documentation task.
- ☐ The matrix is dated and retained with the eight presentation documents.

11. Closing statement

Substantial ownership is distributed across firmware, hardware, backend, dashboard/storage and analytics/safety/quality. Cross-cutting safety semantics remain shared. Source-backed work is described as "the team implemented"; personal authorship is claimed only after the team confirms it.